

Semantic Communication + AI

Can a camera send only the information a receiver needs—without sending every pixel perfectly?



THESIS

Train the sender and receiver together so the information needed for classification survives a limited, noisy link.

A simulation-first capstone with an optional real-radio demonstration later.

What problem are we solving?

Imagine a remote camera sending images to a server that decides what it sees.

REAL-WORLD TENSION

- 1** The sensor is constrained Bandwidth, latency and energy are limited at the edge.
- 2** The wireless link is imperfect Noise can corrupt a short message or make decoding fail.
- 3** The receiver has a narrow goal For classification, the final question is often simply: what is in the image?

DESIGN QUESTION

Why spend the entire link budget reconstructing every pixel if the receiver ultimately needs a task decision?

The project tests this question; it does not assume the learned method must win.

Goal: preserve the receiver's classification decision without requiring a perfect copy of every image.

How the proposed system changes the usual approach

CONVENTIONAL COMMUNICATION Compress the image → protect the bits → reconstruct the file → classify it

Primary objective: recover source bits and pixels

PROPOSED LEARNED LINK Learned sender → noisy link → learned receiver → class decision

Primary objective: preserve information needed for the task

EXPECTED SIGNATURE

As the link worsens, a separated chain may hit a decoding cliff; a learned joint system may degrade more gradually.

IMPORTANT CONTROL

A digital system can also send learned features—so the experiment includes that third arm.

Research question: does the task-oriented joint design help after bandwidth, noise and comparison strength are matched fairly?

Literature survey and research gap

Finite blocklength

[1–4]

Rate backs off from capacity; source and channel dispersions interact.

OPEN ISSUE

Need measured practical short-packet loss.

Learned compression

[11–17]

End-to-end transforms and entropy priors improve rate–distortion.

OPEN ISSUE

Reconstruction optimization is not task optimization.

DeepJSCC

[5–10, 24]

Continuous learned mappings exhibit graceful degradation under mismatch.

OPEN ISSUE

Most image studies foreground reconstruction quality.

Task-oriented communication

[18–23]

Learned representations can preserve remote inference under link constraints.

OPEN ISSUE

Task awareness must be separated from joint coding.

Gap carried forward: separate neural representation learning from joint channel coding in a controlled three-way comparison.

Objectives and research hypotheses

OBJECTIVES

1. Train a residual neural encoder and dual-head decoder through AWGN.
2. Implement a standards-derived, non-strawman JPEG 2000 + NR LDPC baseline.
3. Tune classical quality, code rate and modulation on validation only.
4. Add a shared-encoder digital AI feature control for attribution.
5. Evaluate paired image outcomes once on the sealed test split.
6. Report positive, null or negative outcomes under one frozen protocol.

H1 · PRIMARY

Low-SNR separation: learned – classical_adaptive

A point qualifies when $\sqrt{N} \cdot \Delta(s) / \sigma(s) > 1.96$. H1 is supported only if the longest run R_{obs} contains at least three consecutive qualifying points at or below the training SNR and the calibrated run p-value ≤ 0.05 .

Effect size of record: mean paired accuracy difference over the full low-SNR region.

H2 Graceful vs cliff

Fixed classical curve drops ≥ 30 pp while learned drops ≤ 15 pp over the frozen window.

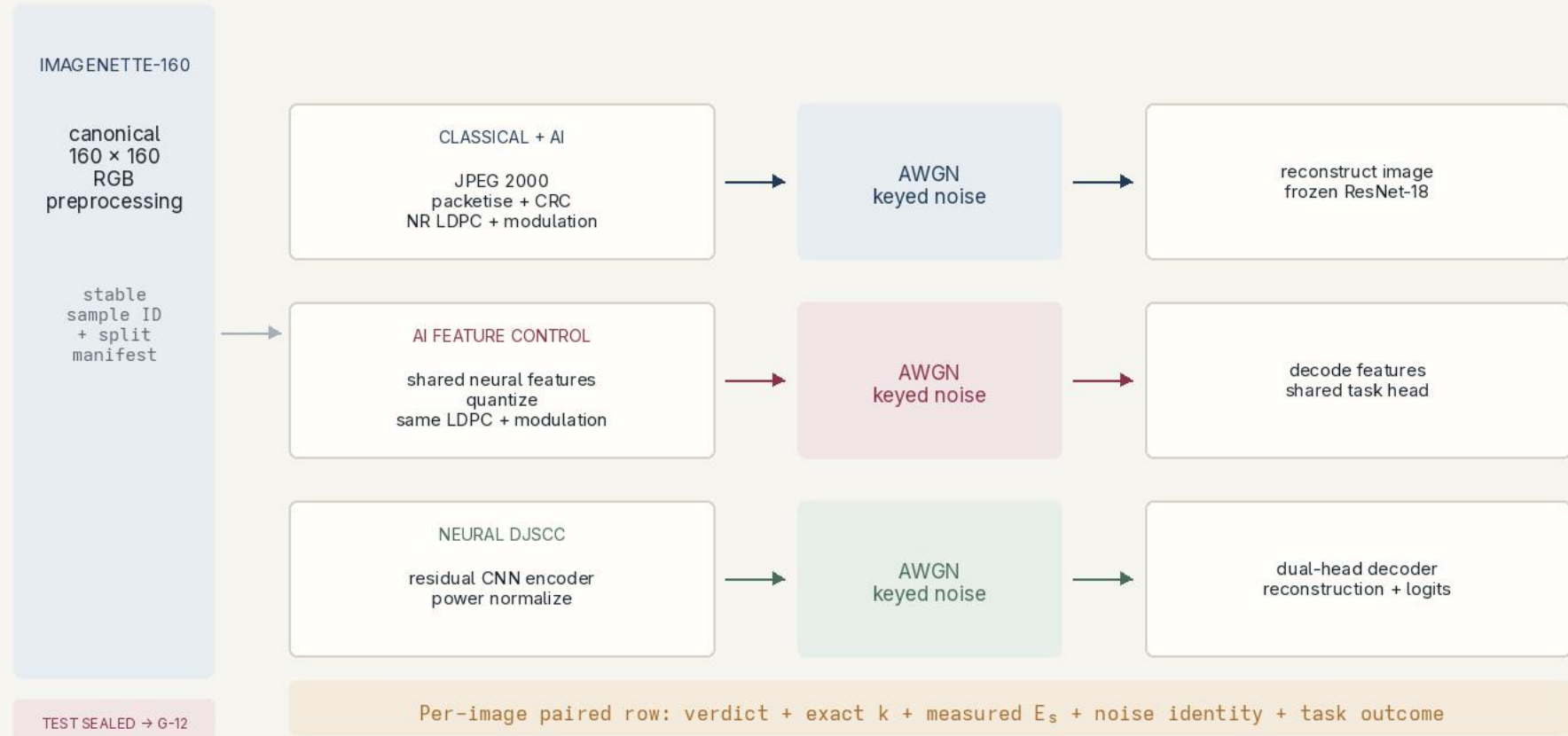
H3 Convergence

Paired gap trends toward zero across SNR; crossover is reported, not required.

H4 Attribution

Credit joint coding only if DJSCC also exceeds the task-aware digital control under the H1 rule.

Proposed methodology: three-system comparison



How we keep the comparison fair

Equal resource

Same complex-symbol budget k and E_s/N_0 definition; realized symbol energy is logged.

Strong comparator

JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM tune per SNR on validation.

Exact accounting

Container, CRC, code-block and filler bytes count; decode failures remain in the denominator.

Measured BLER

Every required physical identity is characterized; missing evidence is never treated as zero BLER.

Outage policy

Constant-class fallback is measured from validation counts—100/1,000—not assumed as 1/10.

Classifier control

Clean and artifact-finetuned classifier passes separate codec shift from link failure.

Attribution arm

Quantized learned features traverse the same digital chain, isolating task awareness from joint coding.

One test campaign

All choices freeze before G-12; paired inference retains complete image × system trajectories.

Fail-closed rule: uncharacterized, infeasible and failed cases are explicit outcomes—not silently dropped rows.

Preliminary work and feasibility

NO HEADLINE COMPARISON EXISTS YET · bounded smoke is integration evidence, not scientific outcome evidence

G-1 · ResNet-18

89.8%

trained from scratch
898/1,000 validation top-1 · PASS

G-7 · neural model

1.64 M

residual DJSCC parameters · batch
32
48.68 s/epoch · 1.004 GiB VRAM

G-2 · digital PHY

≤ 0.0037 dB

observed waterfall displacement
all three modulations within 0.5 dB ·
PASS

W4 · integration

end-to-end

JPEG 2000 → LDPC → AWGN →
decode
records + verifier · bounded PASS

G8_C · COMPLETE

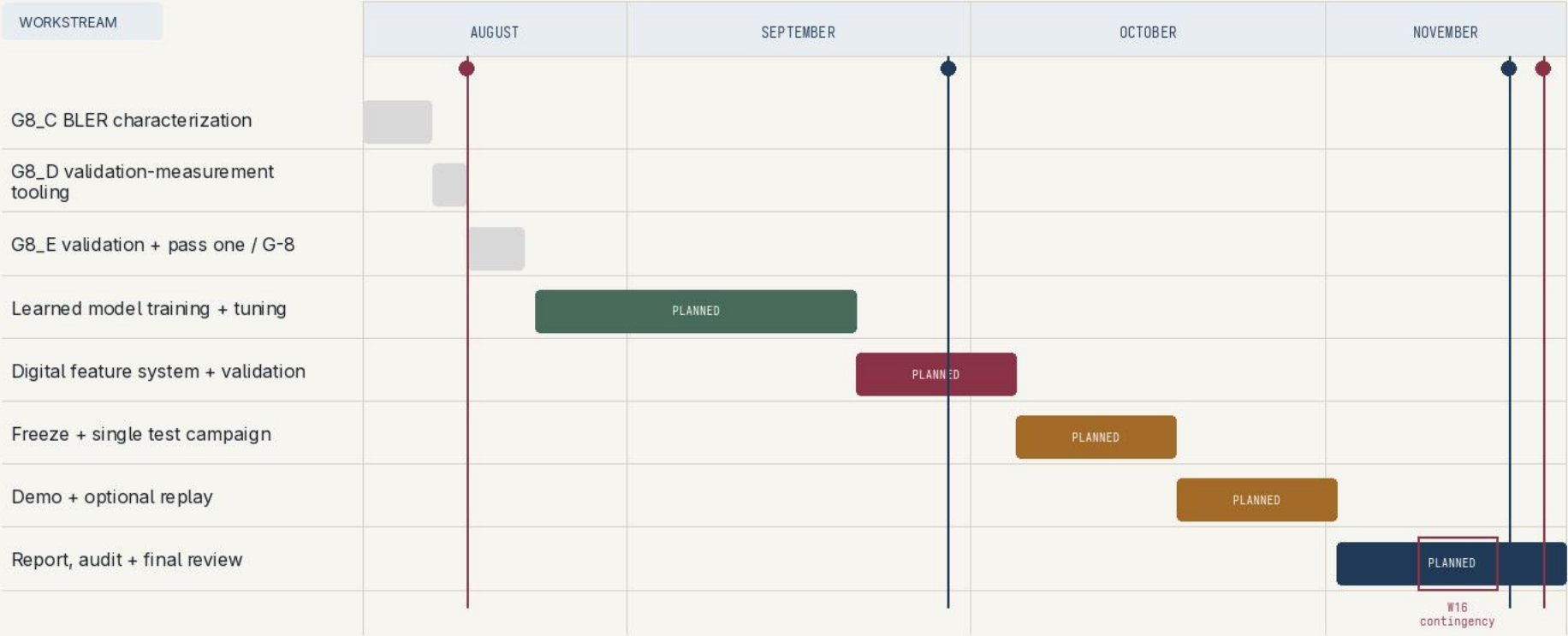
COMPLETE

Pascal successor: 3,213/3,213 identities
153 measured-only curves · successor table frozen

G8_D tooling: COMPLETE
G8_E/E0: PLANNED / NOT STARTED

No full validation measurement/pass one, training/fine-tuning, learned-vs-classical result or test evaluation has occurred.

Project Gantt chart



Review dates: 18-22 Aug · 29 Sep-3 Oct · 17-21 Nov · report due 20 Nov

G8_C and G8_D are complete. G8_E/E0 is next but not started; all later work remains planned.

Edge AI application, AIML contribution and scope

APPLICATION

Edge AI application

Remote cameras and sensors may need receiver-side inference over limited, noisy links.

The sender runs a compact encoder; the receiver owns the decoder and task head. This represents split edge/cloud inference.

The measured task is Imagenette-160 classification, but the method is neural representation learning for communication.

IMPLEMENTATION

Deployment ladder

TIER 1 · required
Simulation-first, offline and reproducible.

TIER 2 · stretch after G-5
Conducted SDR I/Q replay; prerecorded outcome expected.

TIER 3 · stretch
Edge placement only if latency and procurement gates pass.

Candidate: HackRF One + attenuation chain + RTL-SDR; no purchase yet.

METHOD

AI/ML contribution

Residual CNN encoder with GroupNorm and PReLU.

Dual-head decoder for class logits and reconstruction.

Supervised multi-task loss: cross-entropy + λ MSE.

Differentiable AWGN enables end-to-end training.

The digital feature arm isolates representation learning from joint coding.

TECHNICAL BOUNDARY

OpenJPEG 2.5.4 + TS 38.212-derived LDPC/rate matching over abstract AWGN—not a full 5G NR link claim

No guaranteed win, measured energy saving or real-radio generalization; completion means executing the preregistered protocol.

Explicit First Review rubric-to-slide map

CRITERION	SLIDES	EVIDENCE EXPOSED IN THE DECK
Motivation	2–3	Short packets, fixed k , task accuracy, finite-blocklength costs
Objectives	3, 5–7, 10	Neural method, three-arm build, fair protocol and attribution control
Hypothesis	5	Exact H1 run rule; H2–H4 preregistered; crossing not required
Problem Survey	2, 4, 12	30-source synthesis across four literature families
Subject Knowledge	3–8, 10	Residual CNN, multi-task loss, AWGN, controls, evidence and scope
Time Plan	9	Gate-ordered critical path and fixed review/report dates

Final PASS still requires the real PPT review, four-member rehearsal, guide acknowledgement and annotated review-1-basis tag.

Selected references and evidence provenance

SELECTED LITERATURE

[1] Shannon, A Mathematical Theory of Communication, 1948.

[2] Gastpar, Rimoldi & Vetterli, To Code, or Not to Code, 2003.

[3] Polyanskiy, Poor & Verdú, Finite Blocklength, 2010.

[4] Kostina & Verdú, Lossy JSCC at Finite Blocklength, 2013.

[5] Boursoulatz, Kurka & Gündüz, DeepJSCC, 2019.

[6] Kurka & Gündüz, DeepJSCC-f, 2020.

[7] Kurka & Gündüz, Bandwidth-Agile DeepJSCC, 2021.

[11] Ballé, Laparra & Simoncelli, Learned Compression, 2017.

[12] Ballé et al., Scale Hyperprior, 2018.

[17] Blau & Michaeli, Perception-Distortion, 2018.

[18] Jankowski, Gündüz & Mikołajczyk, Wireless Retrieval, 2021.

[19] Shao, Mao & Zhang, Task-Oriented Edge Inference, 2021.

[20] Xie et al., Deep Learning Enabled Semantic Communication, 2021.

REPOSITORY EVIDENCE

G-1	results/reference_classifier/g1_adjudication.json 89.8% validation; test sealed
G-7	results/profiling/g7_djscc_profile.json 1.64M params; measured epoch/VRAM
G-2	results/baseline/g2/g2_adjudication.json golden vectors + BLER conformance
W4	results/baseline/w4/integration_adjudication.json bounded end-to-end integration
G8_C	results/baseline/g8_pascal_successor/successor_bler_table.json 153 measured-only curves; successor table frozen
G8_D	results/baseline/g8_d/d7_handoff.json D0-D7 GREEN; tooling ready; full campaign not started
Spec	spec/SPEC.md normative hypotheses and requirements
Plan	docs/gantt-plan.md dates, gates and current status
G-1	docs/deployment_decisions.md

Refresh G8_C/G8_D status from the handoff files on submission day.